SUPERNOVA SMARTHOME

ESP8266 WiFi Smart Home Automation System

User Manual & Installation Guide

Web App | Amazon Alexa | Telegram Bot | Physical / Touch Switches | Scheduled Timers

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1. Introduction & Features

SUPERNOVA SMARTHOME is a WiFi-based home automation firmware for the ESP8266 microcontroller. It allows you to control up to 8 relay channels from a mobile-friendly web dashboard, Amazon Alexa, Telegram, or physical/touch switches — with scheduling, real-time synchronization across devices, and wireless firmware updates.

Key Features

- Responsive, login-protected web dashboard with real-time updates over WebSocket
- 5-Channel (GPIO) or 8-Channel (I2C / PCF8574) hardware modes — switchable in software
- Physical switch support: Manual (maintained toggle) or Touch (TTP223 momentary) modes
- Duration-based timers (ON/OFF after X minutes) and Date/Time scheduled ON/OFF, both survive power cuts
- Configurable power-on behavior per switch: restore last state / always ON / always OFF
- Telegram Bot control with an inline-keyboard switch panel and per-user access control
- Amazon Alexa voice control (Philips Hue device emulation)
- WiFi Access Point + Station modes, with optional Static IP configuration
- Over-the-Air (OTA) firmware updates
- Session-based web authentication with secure cookie login
- QR-code WiFi sharing, custom switch names & icons, light/dark themes

2. Safety Warning

This project is capable of switching mains-voltage relays. Incorrect wiring of mains electricity can cause fire, electric shock, or death.

- Always disconnect mains power before wiring or servicing any relay module.
- If you are not experienced with electrical wiring, consult a qualified electrician.
- Use relay modules and enclosures rated for the voltage/current of your load.
- The author and contributors are not responsible for any damage, injury, or loss arising from the use of this project.

3. Hardware Requirements

Component	Notes
ESP8266 development board	NodeMCU / Wemos D1 Mini or similar
Relay module(s)	5-channel (GPIO mode) or 8-channel (I2C mode)
PCF8574 I/O Expander	Only required for 8-Channel (I2C) mode
Push / toggle switches	For manual physical control
TTP223 touch sensor(s)	Optional — for touch-switch mode

4. Pin Mapping

5-Channel GPIO Mode

Switch	Relay Pin	Input Pin
1	GPIO5 (D1)	GPIO16 (D0)
2	GPIO4 (D2)	GPIO1 (TX)
3	GPIO14 (D5)	GPIO3 (RX)
4	GPIO12 (D6)	GPIO2 (D4)
5	GPIO13 (D7)	GPIO15 (D8)

8-Channel I2C Mode (via PCF8574, default address 0x27)

Switch	Output	Input Pin
1	PCF8574 P0	GPIO16 (D0)
2	PCF8574 P1	GPIO14 (D5)
3	PCF8574 P2	GPIO12 (D6)
4	PCF8574 P3	GPIO13 (D7)
5	PCF8574 P4	GPIO2 (D4)
6	PCF8574 P5	GPIO15 (D8)
7	PCF8574 P6	GPIO3 (RX)
8	PCF8574 P7	GPIO1 (TX)

I2C bus: SDA = GPIO4 (D2), SCL = GPIO5 (D1).

Note: GPIO1 (TX) and GPIO3 (RX) are used as switch inputs on certain channels. When these are active, the onboard Serial interface is automatically disabled by the firmware to prevent conflicts. GPIO2 sharing with the onboard status LED is also auto-detected and handled.

5. Required Arduino Libraries

Install the following via Arduino IDE → Library Manager (or manually from GitHub). Also install the **ESP8266 board package** via Boards Manager (URL: https://arduino.esp8266.com/stable/package_esp8266com_index.json).

Library	Purpose
ESP8266WiFi / ESP8266WebServer / DNSServer / EEPROM / Wire / ArduinoOTA / WiFiClientSecure	Bundled with the ESP8266 board package
WebSocketsServer (Links2004/arduinoWebSockets)	Real-time UI updates
fauxmoESP (vintlabs/fauxmoESP)	Amazon Alexa emulation
UniversalTelegramBot (witnessmenow)	Telegram Bot integration

6. Firmware Installation

- Open the .ino file in Arduino IDE.
- Select your ESP8266 board under Tools → Board.
- Install all libraries listed in Section 5.
- Connect the board via USB and select the correct Port.
- Click Upload.

7. First-Time Setup (Access Point Mode)

On first boot, the device starts its own WiFi Access Point so you can configure it before it has a home network to join.

Setting	Default Value
WiFi SSID	SUPERNOVA SMARTHOME
WiFi Password	12345678
Device IP Address	192.168.4.1
Web Login Username	admin
Web Login Password	admin

Change the default WiFi password and web login credentials immediately from the Settings page after your first login.

8. Connecting to Your Home WiFi

Go to Settings → WiFi, enter your home network's SSID and password, and switch the device to Station (STA) mode. You may optionally enable a Static IP by providing an IP address, gateway, and subnet mask — otherwise the device will use DHCP.

After saving WiFi settings the device restarts automatically and reconnects using the new configuration.

9. Web Dashboard Guide

After logging in, the dashboard shows a tile for each switch with an ON/OFF toggle, custom name, and icon. State updates are pushed instantly to every connected browser over WebSocket, so multiple people viewing the dashboard always stay in sync.

The Settings area is organized into sections:

- **WiFi** — Access Point / Station mode, credentials, Static IP
- **Switch Names & Icons** — rename each channel and pick an icon
- **Timers** — duration and date/time based scheduling per switch
- **Power-On Default** — behavior after a reboot or power cut
- **Telegram** — bot token, authorized chat IDs, enable/disable
- **Alexa** — enable/disable voice control
- **Security** — web login username/password
- **Hardware Mode** — 5-Channel (GPIO) or 8-Channel (I2C)
- **Physical Switch Mode** — Manual or Touch
- **OTA** — enable/disable wireless firmware updates

10. Physical & Touch Switch Modes

Manual Switch Mode

For maintained toggle switches. Any change in the pin's state (HIGH↔LOW) toggles the corresponding relay.

Touch Switch Mode

For TTP223-style momentary touch sensors. Only a LOW-to-HIGH transition (a single tap) toggles the relay, matching the momentary pulse behavior of a touch sensor.

Both modes share a 30 ms debounce filter to reject electrical noise/bounce.

11. Timers & Scheduling

Two independent timer types are available per switch:

- **Duration Timer** — turn a switch ON or OFF automatically after a set number of minutes from now.
- **Date/Time Timer** — schedule a switch to turn ON or OFF at a specific future date and time.

Setting one timer type for a switch automatically cancels the other type for that same switch, avoiding conflicting schedules. Scheduled Date/Time timers are stored in EEPROM and survive a power cut or reboot. Manually toggling a switch (from the web app, Telegram, Alexa, or a physical switch) cancels any active timer for that switch.

12. Power-On Behavior Settings

Each switch can be configured independently for what happens right after the device boots (e.g. following a power cut):

Mode	Behavior
Last State	Restores whatever state the switch was in before power was lost (default)
Always ON	Switch turns ON every time the device boots
Always OFF	Switch turns OFF every time the device boots

13. Telegram Bot Setup & Commands

Setup Steps

- Create a bot via @BotFather on Telegram and obtain your Bot Token.
- Get your numeric Chat ID (e.g. via @userinfobot).
- In the Web UI, go to Settings → Telegram, enter the token and up to 3 authorized Chat IDs, then enable the bot.
- Only the configured Chat IDs may control the system; unauthorized access attempts trigger an alert message to the authorized admin chat IDs.

Command Reference

Command	Description
/menu	Interactive inline switch control panel
/status	Current state of all switches
/on <n> /off <n>	Turn switch n on / off
/allon /alloff	Turn all switches on / off
/timeron <n> <YYYY-MM-DD HH:MM>	Schedule switch n to turn ON at date/time
/timeroff <n> <YYYY-MM-DD HH:MM>	Schedule switch n to turn OFF at date/time
/cancelon <n> /canceloff <n>	Cancel a scheduled timer
/timers	List all active timers
/uptime	Show device uptime
/restart	Restart the device
/botoff	Disable the Telegram bot remotely
/help	Show all available commands

14. Amazon Alexa Setup

- Enable Alexa from the Web UI Settings page.
- Ask your Echo/Alexa device to "discover devices" (Smart Home section of the Alexa app).
- Each configured switch will appear to Alexa as a separate device, named after its custom switch name.
- Voice commands such as "Alexa, turn on [switch name]" will then control the corresponding relay.

15. OTA Firmware Updates

Once the device is connected to your home WiFi and OTA is enabled in Settings, new firmware can be uploaded wirelessly from Arduino IDE via Tools → Port → (select the device's network port) instead of a USB cable.

16. Factory Reset

If you are locked out (forgotten WiFi/web password, or the device cannot join your network), a hardware factory reset is available:

- Locate the reset button wired to GPIO0 (D3) on the board.
- Press and hold the button. The status LED will blink rapidly while held.
- Keep holding for 5 seconds — the LED will flash to confirm, and the device will erase all saved settings and restart.
- After restart, the device returns to factory defaults (see Section 7) and boots back into Access Point mode.

This action is irreversible and erases WiFi credentials, switch names, timers, Telegram/Alexa settings, and the web login. Reconfigure the device from scratch afterward.

17. Troubleshooting / FAQ

The device is not visible as a WiFi network.

Ensure it is powered and has not already joined your home network in Station mode. Check the router's client list, or perform a factory reset (Section 16) to force it back into Access Point mode.

I forgot my web login or WiFi password.

Perform a hardware factory reset (Section 16) and reconfigure the device from the default Access Point credentials.

Physical switch on Switch 2/3 (5-ch mode) or Switch 7/8 (8-ch mode) behaves oddly.

These channels share pins with the onboard Serial (TX/RX) interface. Make sure nothing else is connected to those pins that could interfere, and confirm the correct Hardware Mode and Physical Switch Mode are selected in Settings.

Alexa cannot discover my devices.

Confirm Alexa is enabled in Settings, the device and your Echo are on the same WiFi network, and re-run "discover devices" from the Alexa app.

Telegram bot does not respond.

Verify the bot token is correct, the bot is enabled in Settings, and that you are messaging from one of the configured authorized Chat IDs.

Static IP does not seem to apply.

Double-check that the IP, Gateway, and Subnet Mask are all valid and on the same subnet as your router, then save settings again — the device restarts to apply network changes.

18. Disclaimer & License

This project is provided as-is, for educational and hobbyist use. It involves controlling electrical relays which may switch mains voltage. Use at your own risk. Always follow local electrical safety codes and disconnect power before wiring.

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